

# MOHD HAMZA KHAN

## AI Engineer

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## PROFESSIONAL SUMMARY

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Early-career AI Engineer with hands-on experience building production-grade RAG systems, LLM optimization pipelines, and agentic AI workflows. Proven ability to deliver end-to-end AI solutions from prototype to deployment, with a strong focus on performance, data privacy, and scalability. Proficient in Python, LangChain, vector databases, and prompt engineering.

## EXPERIENCE

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### AI Winter Intern | Mirai School of Technology

Jan 2025 – Feb 2025 • Remote

- Developed production-ready RAG applications using semantic search with vector embeddings and FAISS, enabling context-aware document retrieval and response generation.
- Integrated OpenAI LLM APIs into production workflows with advanced prompt optimization, token management, and hallucination mitigation strategies.
- Implemented AI automation workflows for real-world problem solving using Generative AI frameworks including LangChain and Agentic pipelines.
- Built Retrieval-Augmented Generation (RAG) systems leveraging embeddings and vector databases to generate accurate, context-aware responses.
- Collaborated with cross-functional teams in agile sprints and contributed to code reviews and technical documentation.

### Web Development Intern | SafalUday Foundation

Apr 2025 – Jul 2025 • Remote

- Developed responsive, production-ready user interfaces using HTML, CSS, JavaScript, Bootstrap, and React with accessibility best practices.
- Implemented backend logic with PHP for form handling, data validation, and secure database operations (MySQL).
- Integrated and tested RESTful APIs and third-party services using Postman; documented API specs for team collaboration.
- Participated in agile ceremonies (standups, sprint planning) and maintained technical documentation for knowledge sharing.

## PROJECTS

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### Local AI-Powered PDF QA System (RAG) | Mar 2026 – Present • Lucknow, India

- Engineered an end-to-end RAG system for document question-answering with 100% local execution — eliminating API costs and ensuring data privacy.
- Implemented semantic search using sentence-transformers with FAISS vector database for O(1) similarity search performance.
- Integrated Phi-3 Mini via Ollama for fully offline LLM inference, achieving sub-100ms query latency.
- Engineered efficient text chunking (recursive, overlap-based) and document indexing to optimize retrieval accuracy and latency.
- Deployed on local machine using Docker for reproducibility and scalability; achieved 95%+ retrieval accuracy on test documents.

### Convolve 3.0: AI/ML Hackathon — Round 2 Qualifier | Jan 2025 • Lucknow, India

- Built an ML model to predict optimal email send-time for IDFC First Bank customers, analyzing historical interaction data.
- Engineered features including time-of-day, day-of-week, and customer segments; achieved 85%+ accuracy using scikit-learn with cross-validation and hyperparameter tuning.
- Qualified for Round 2 of pan-IIT hackathon, demonstrating ability to solve real-world business problems under tight time constraints.

## EDUCATION

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**B.Tech in Computer Science** | Integral University, Lucknow, UP

*Aug 2022 – Jul 2026*

**Intermediate – Diploma in Electronics Engineering** | University Polytechnic, A.M.U., Aligarh, UP

*Aug 2019 – Jul 2022*

## TECHNICAL SKILLS

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**AI/ML & LLM:** RAG, Prompt Engineering, LLM Optimization, LangChain, Agentic AI Workflows, LLM Orchestration, Vector Databases (FAISS, Pinecone, Chroma), Local LLMs (Ollama, Phi-3), Evaluation Metrics (BLEU, ROUGE, BERTScore)

**Languages & Frameworks:** Python, Django, PHP, Laravel, JavaScript, React, Tailwind CSS, HTML/CSS, SQL, MongoDB, REST APIs

**Tools & Platforms:** Postman, Docker, Git, Jupyter Notebooks

## ACHIEVEMENTS

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- Finalist at Vibe Singularity Hackathon (RunAnywhere YC W26) — Built a privacy-first Local AI Study Buddy with 100% local LLM execution (Phi-3 Mini via Ollama) in 48 hours, eliminating API costs.
- Round 2 Qualifier – Convoke 3.0 AI/ML Hackathon (pan-IIT, IDFC First Bank) — Developed predictive ML model for optimal email send-times with 85%+ accuracy using feature engineering.
- 2nd Place – National Hackathon — Led team to runner-up finish in a fast-paced, collaborative AI development environment.
- Increased user engagement by 25% through data-driven UX improvements and iterative product enhancements.

## CERTIFICATIONS & CONTINUOUS LEARNING

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**Actively learning:** LangGraph, Autogen, Advanced LoRA Fine-Tuning, and Production Deployment Patterns for Agentic AI Systems.

## INTERESTS

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Football, Table Tennis, Badminton • AI Research & Open-Source Contributions